# Network Security

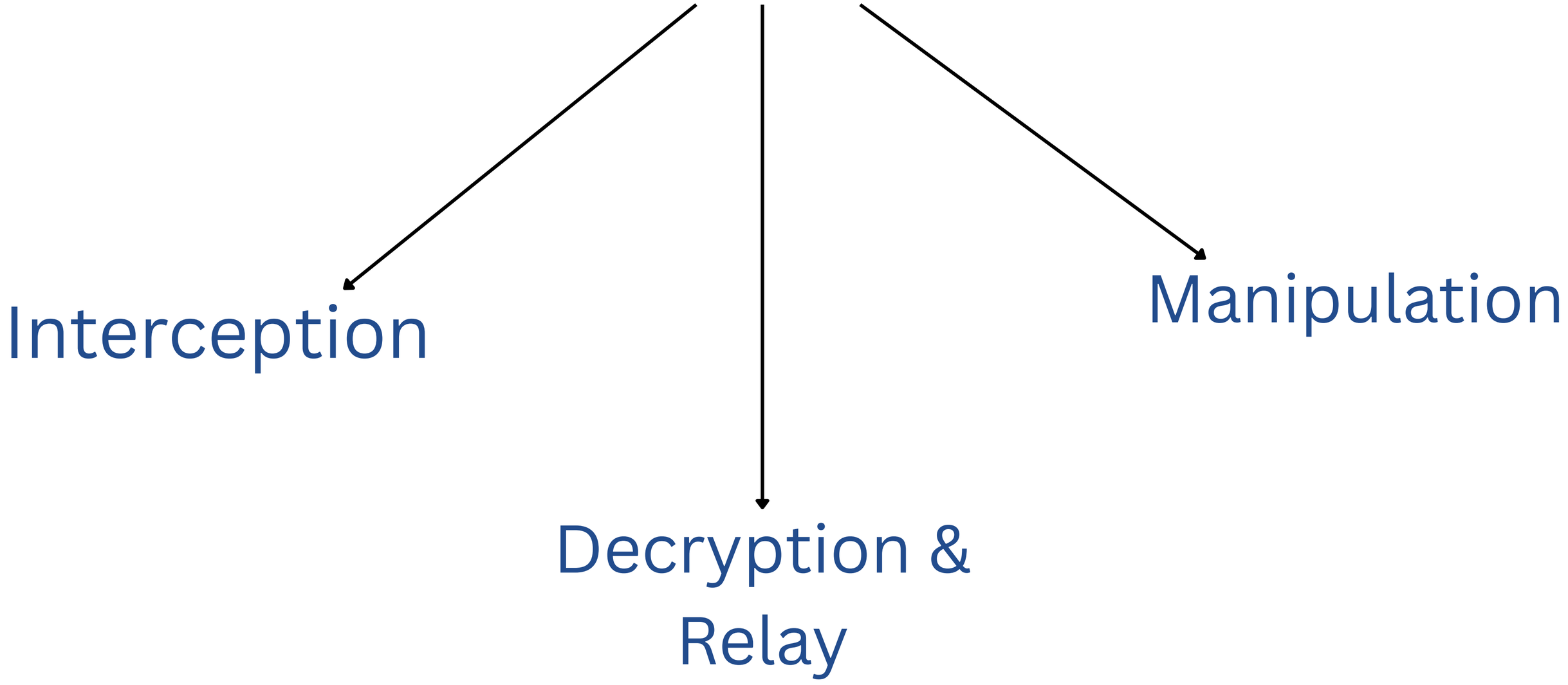
# Group 02

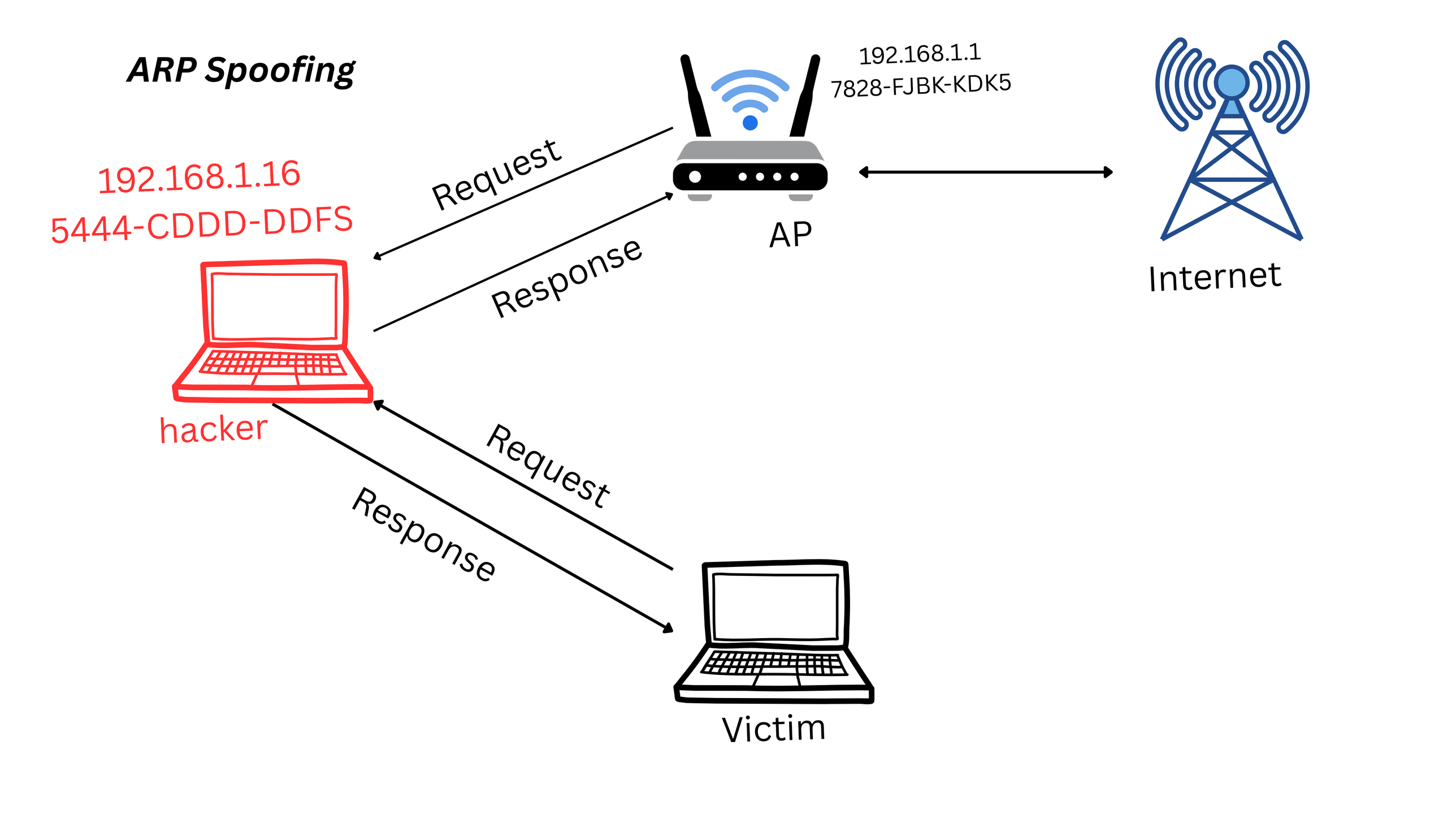
# Man-in-the-Middle Attack


- SOK Kimheng
- TEK Chansetha
- Soeng Prakbormey

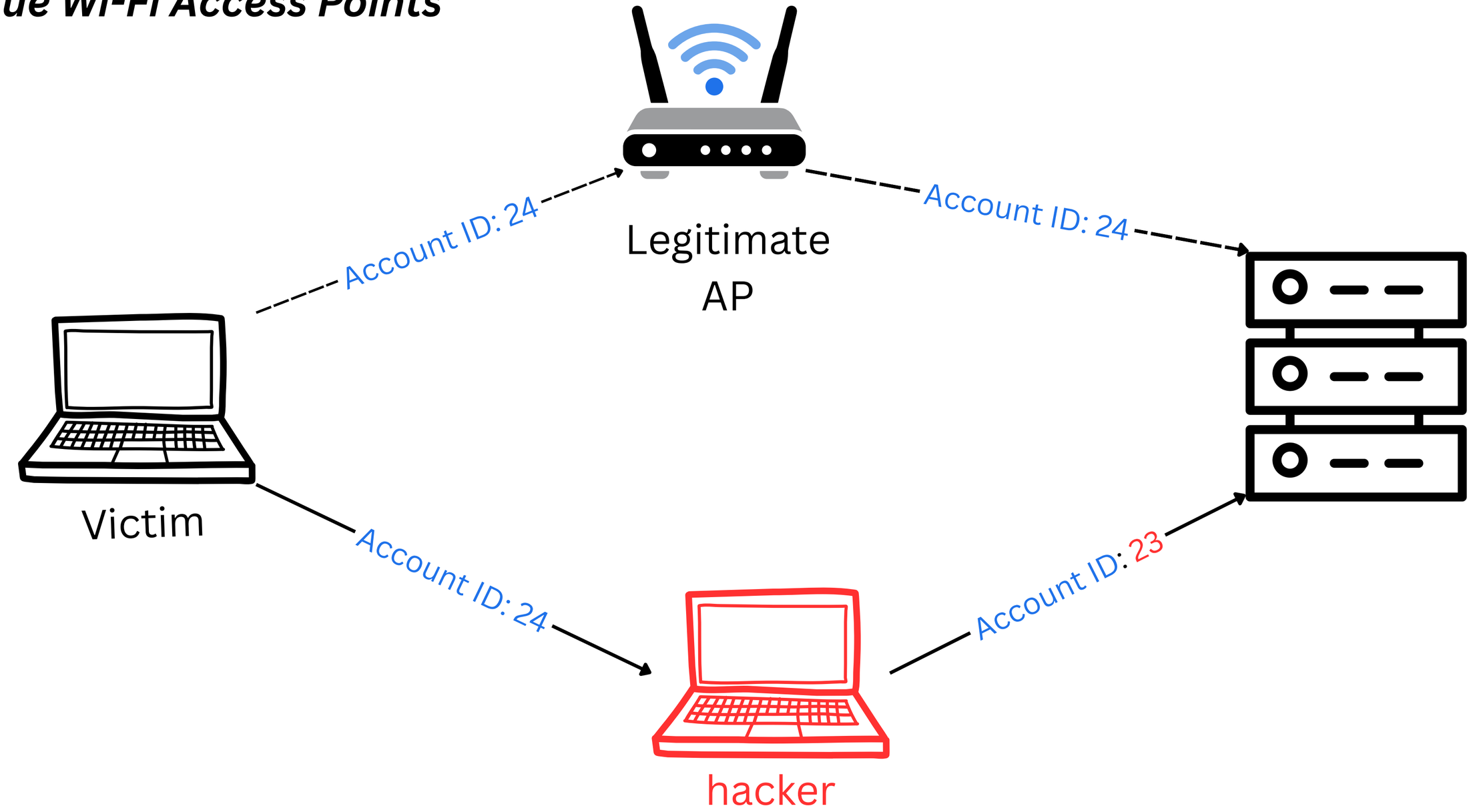
Lecturer: M. ANG Ousa

# Man-in-the-Middle Attack

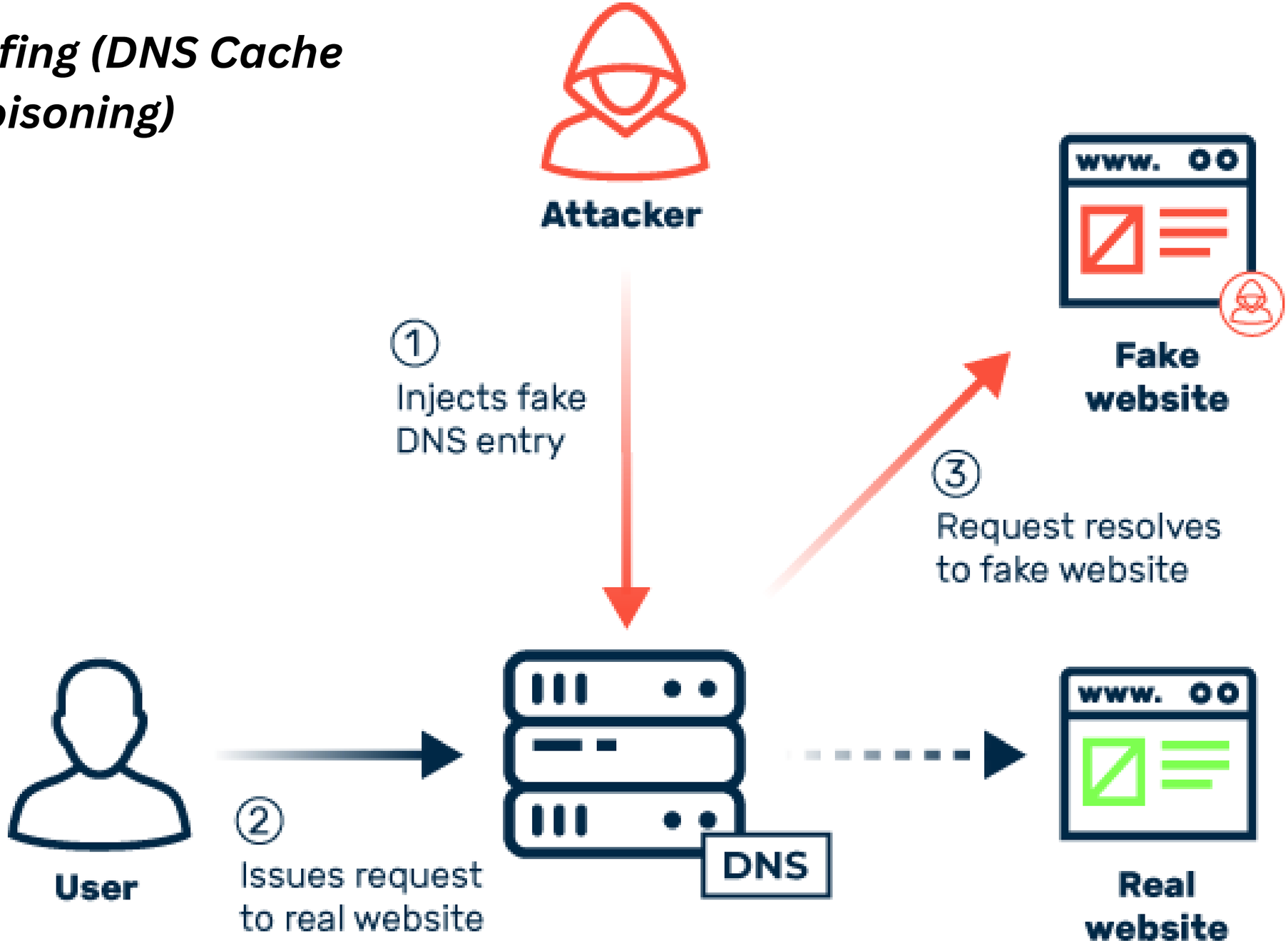



ARP Spoofing
192.168.1.16
5444-CDDD-DDFS
hacker
Request
Response
192.168.1.1
7828-FJBK-KDK5
AP
Internet
Request
Response
Victim

***Rogue Wi-Fi Access Points***

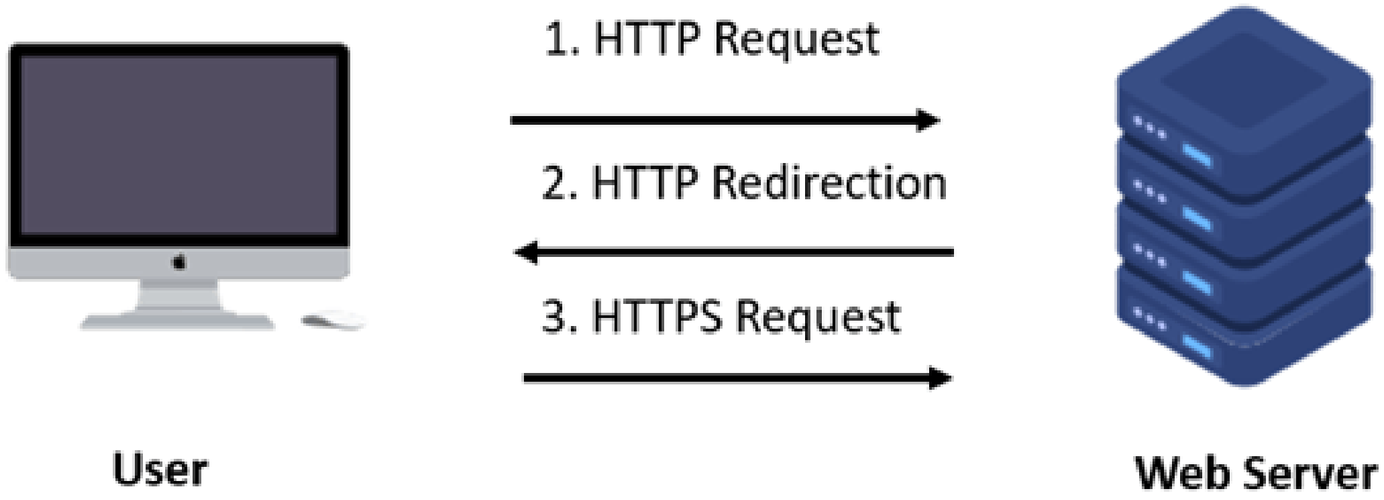


# *DNS Spoofing (DNS Cache Poisoning)*



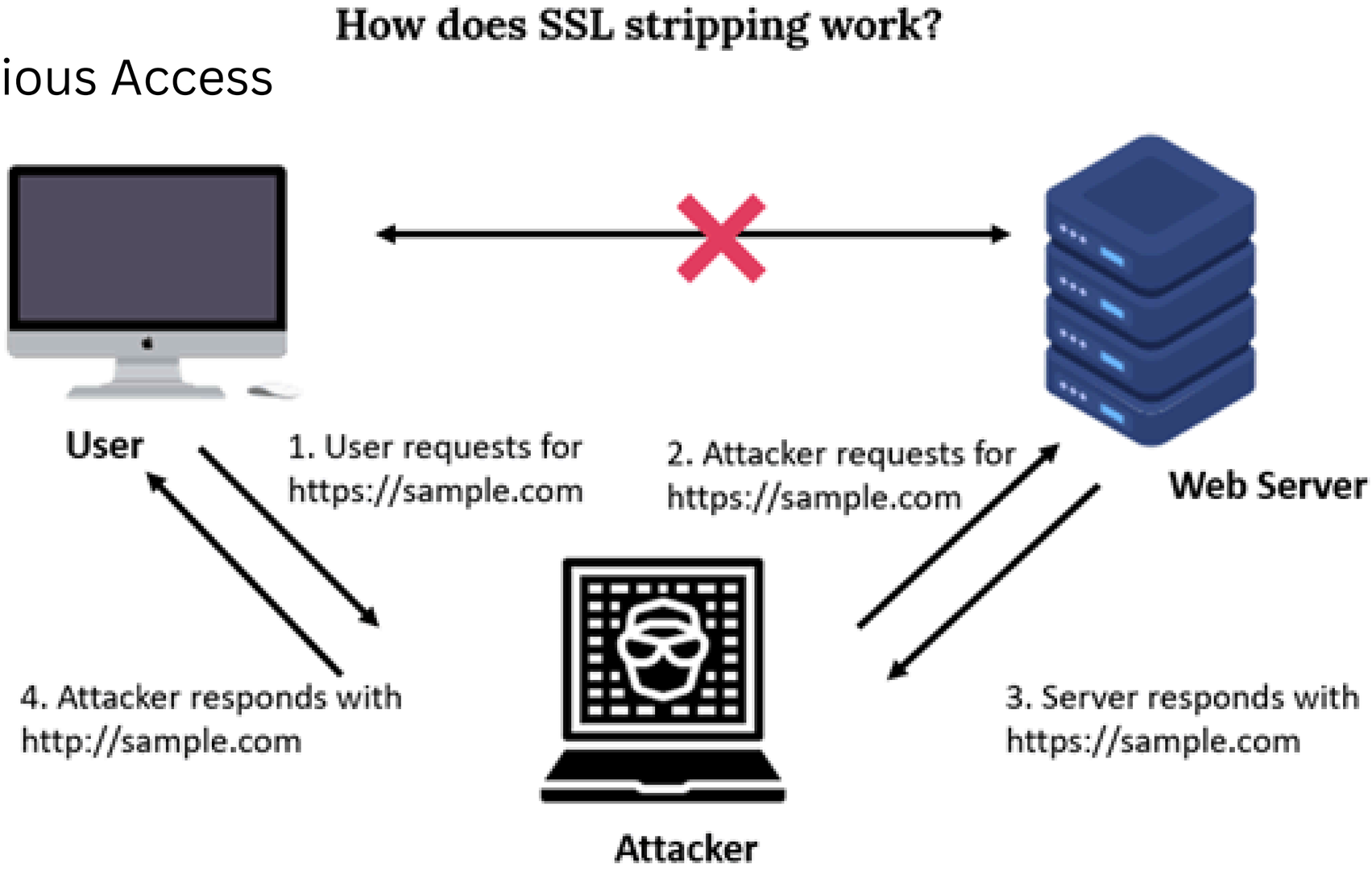
# *SSL/TLS Stripping*

Normal Access

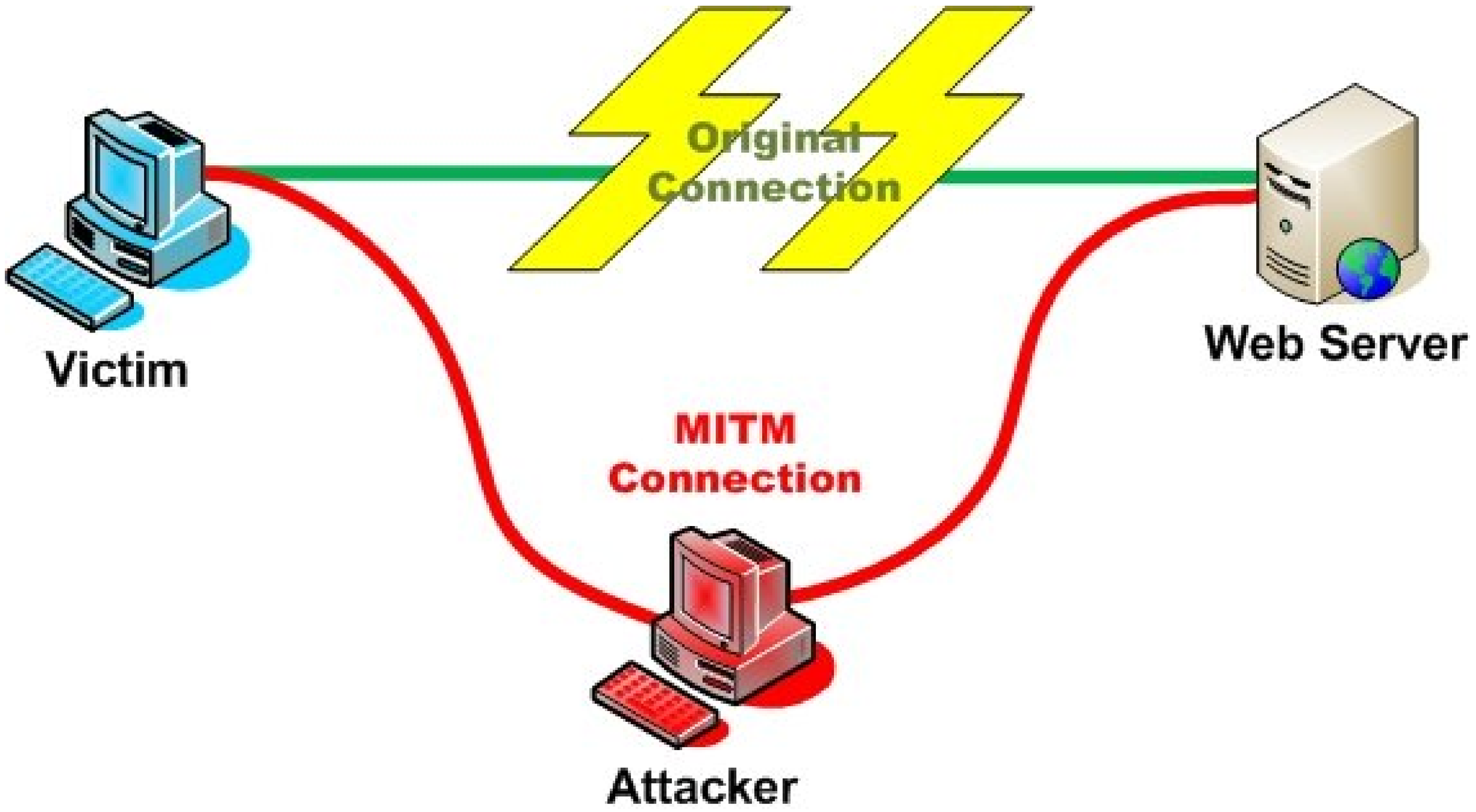


# *SSL/TLS Stripping*

Malicious Access



## *HTTPS Spoofing*



# The 2011 DigiNotar

Netherlands and Iran

***What happened***

## The attack

Attackers breached the DigiNotar certificate authority and issued fraudulent certificates, including one for *.google.com.

## The interception

Traffic was redirected so browsers saw a trusted certificate while the attacker read Gmail sessions. Around 300,000 users, mostly in Iran, were exposed.

## The Outcome

Browsers removed trust in DigiNotar and the company went bankrupt within months.

# Root Causes and Lessons

### Patching

The Struts patch was available in March 2017 but never applied.

### Segmentation

Weak segmentation and plaintext credentials let attackers move freely.

### Visibility

An expired certificate left outbound traffic uninspected for months.

*Takeaway: a basic hygiene and visibility failure, not an advanced technique.*

## *Prevention Strategies*

**For Users:**

- Avoid executing sensitive transactions (like banking or shopping) on unsecured public Wi-Fi networks.
- Always verify that the website URL begins with HTTPS and displays a valid lock icon.
- Utilize a Virtual Private Network (VPN) to encrypt all network traffic leaving your device.
- Deploy robust Multi-Factor Authentication (MFA). Note that traditional MITM attacks cannot easily bypass MFA, though specialized evolutions like Adversary-in-the-Middle (AITM) use reverse-proxies to target session cookies directly.

**For Organizations:**

- Enforce comprehensive, site-wide encryption protocols via SSL/TLS.
- Implement HTTP Strict Transport Security (HSTS) policies to ensure browsers never fall back to insecure HTTP connections.

# References:


- sosafe-awareness.com
- imperva.com
- cyberlaw.ccdcoe.org/wiki/DigiNotar_(2011)
- gemini.google.com